Licensed Electrician's Theory (LET) Licence Assessment

Sample Paper Three (2021)

Candidate Surname	
Candidate Given Names	

Reference Material

- AS/NZS 3000:2018 Wiring Rules
- AS/NZS 3012:2019 Electrical installations – Demolition and Construction sites
- AS/NZS 3008.1.1:2017 Electrical installations – Selection of cables
- Electricity Safety (General) Regulations 2019
- AS/NZS 4836:2011 Safe working on or near low voltage electrical installations and equipment

Instructions

- Personal notepads and paper are not permitted.
- Pens only must be used. Answers in pencil may not be marked.
- Do not remove any sheets from this assessment paper or the room.
- Papers with no name or signature will not be marked.
- Units must be shown to obtain full marks.

Results

Candidates need to obtain 75% or more to pass this assessment. If a mark of 74% or less is achieved, a minimum of 14 days is required before you are permitted to re attempt the assessment.

I, the above named candidate confirm:

- I understand the instructions provided to me.
- do not have any unauthorised materials in my possession
- have not attempted the Licensed Electrician's Theory Assessment at any venue within the past 14 days.

Candidate			
	Print name	Signature	Date

Working Time: 2 hours and 15 minutes

At the end of this time you will be asked to stop.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Total
Mark																			

Final Percentage	Pass/Fail

Supervisor			
	Print name	Signature	Date
Assessor			
	Print name	Signature	Date
Reviewed by (If necessary)			
	Print name	Signature	Date

AS/NZS 3000 WIRING RULES

In the following **four** Wiring Rules questions:

- You are required to write the Wiring Rules Clause and/or Table number in the space provided.
- The correct Wiring Rules Clause and Subclause must be given, eg. (3.5.2 (b) (i))

The correct answer to both parts must be given to obtain full marks.

Question 1. State the name of the point at which the consumers mains or underground service cable enters a structure.

Wiring Rules Clause Number:

[2 + 2 = 4 Marks]

Question 2. State two requirements that apply to socket outlets used in Separated Extra Low Voltage (SELV) and Protected Extra Low Voltage (PELV) installations.

1)

2)

Wiring Rules Clause Number:

[1 + 1 + 2 = 4 Marks]

Question 3. Complete the following sentence:
"A switch used for the isolation of a fixed or stationary cooking appliance should be located within ....metres of the appliance."

Wiring Rules Clause Number:

[2 + 2 = 4 Marks]

Question 4. State the condition in which festoon lighting lamps may be installed within arm's reach.

Wiring Rules Clause Number:

[2 + 2 = 4 Marks]

AS/NZS 3012 CONSTRUCTION AND DEMOLITION SITES

In the following **two** AS/NZS 3012 questions:

- You are required to write the Standard's Clause number and/or Table number in the space provided.
- The correct Clause and Sub-clause number must be given. e.g. 2.10.2 (f).

The correct answer to both parts must be given to obtain full marks.

Question 5. What requirement must be met to enable Festoon lighting to be installed on a wall 2.3m above the bottom of a lift shaft?

Standard Clause Number: [2 + 2 = 4 Marks]

Question 6. State the recommended minimum lighting levels for general areas on a construction site.

Standard Clause Number: [2 + 2 = 4 Marks]

ELECTRICITY SAFETY (GENERAL) REGULATIONS 2019

In the following Regulation question, you are required to:

- write your answers on the line/s below each question
- write the complete Regulation and Sub-Regulation number, if applicable, in the space provided, e.g. 401(e)(3).

The correct answer to both parts must be given to obtain full marks.

Question 7. Give two (2) examples where a multiple earth neutral connection must be installed in an electrical installation required to be earthed.

1)

2)

Regulation Number: ----------------------------------- [1 + 1 + 2 = 4 Marks]

ELECTRIC SHOCK SURVIVAL

Electric Shock Survival

DANGERS
Check for your own safety and the safety of the casualty and bystanders.
HIGH VOLTAGE-Wait until the power is turned off.
LOW VOLTAGE-immediately switch off the power. If this is not practicable, pull or push the casualty clear of the electrical contact using material, such as wood, rope, clothing, plastic or rubber. Do not use metal or anything moist.

RESPONSIVENESS
Check for response (verbal and tactile stimuli), touch and talk.

SEND/Shout FOR HELP
Send a bystander to DIAL 000 Ambulance
If available send for Automatic External Defibrillator (AED)
If alone shout for help.

AIRWAY Place the casualty on his/her back.
Tilt the head back and raise the chin forward.

BREATHING Check for normal breathing, observe chest movement, listen and feel for breathing.
Give two initial breaths.
In the absence of normal breathing, if no one has gone for help, place casualty in recovery position and go for help.

CIRCULATION
Position hands on centre of the chest.
Give 30 chest compressions followed by 2 breaths. Depress breastbone 1 / 3 the chest depth (approx. 4cm or 5cm) at the rate of 100 compressions a minute.
As soon as available attach AED and follow its instructions.
Continue CPR" 30 compressions: 2 breaths.
When casualty is normal breathing returns cease resuscitation and move the casualty into the **recovery** or **coma** position.
Keep a constant watch on the casualty, to ensure that they do not stop breathing again, until trained assistants take over.

Question 8. State how an electric shock victim is to be placed to open their airways when applying Cardio Pulmonary Resuscitation

1.

2.

[2 + 2 = 4 Marks]

CABLE SELECTION

Question 9.
Two 35 mm2 three core X90 insulated, sheathed and armoured copper cables, including earthing conductors, are connected in parallel to supply a three phase load.

The cables are protected by a circuit breaker and installed touching directly in the ground at a depth of 2.5m.

(i) Neglecting voltage drop, what is the TOTAL maximum current carrying capacity of the cables which form the circuit?

(ii) If using HRC fuses to protect this circuit, the current rating of the fuse should not exceed what value?

All calculations including the final answer must be completed to a maximum of **two decimal places.**

Table details and units must be shown below to obtain full marks.

Part (i)

	Answer		Answer		
Table 3 (______?)		Item			
Table		Column			Answer
Derate/rating table		Column		Factor	
Derate/rating table		Column		Factor	

Current Carrying Capacity Answer

Part (i) Answer:	

Question 9. Part (ii)

	Answer		Answer		Answer
Derate/rating table		Column		Factor	

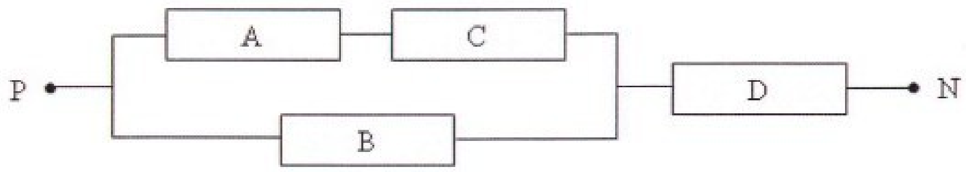
HRC Fuse Size Answer

Part (ii) Answer:	

[1 + 2 + 1 + 1 + 2 + 1 = 8 Marks]

DC CIRCUITS

Question 10.



The following values apply to the diagram above:

The total voltage across points P and N = 200V

A is 50Ω; B is 140Ω; C is 90Ω and D is 30Ω

Calculate:

(i) the current flowing through resistor C
(ii) the total voltage across resistor A
(iii) the total power dissipated in resistor B

All calculations and units must be shown to obtain full marks to a maximum of **two decimal places.**

(i) Current:		**(ii) Voltage:**		**(iii) Power:**	

[2 + 2 + 2 = 6 Marks]

MAXIMUM DEMAND

Question 11.
Calculate the Maximum Demand of a 230V main supplying a Factory (garage / workshop) distribution board.

The load connected to the switchboard is:-

* 4 circuits each of 10 - 10A socket outlets
* 2 circuit of 2 - 15A socket outlets
* 3 circuits each of 8 - incandescent lighting points
* 1 circuit supplying a petrol pump (4 Amp motor)
* 1 circuit supplying an arc welder with a primary current rated at 25 amps

All calculations including the final answer must be completed to a maximum of **two decimal places.**

All relevant table details, including table, column and load groups used.
Calculations and units must be shown to obtain full marks.

Table		Column	

Equipment	Load Group	Calculation	Maximum Demand
		Total Maximum Demand:	

[1+ 2 + 1 + 1 + 1 + 1 + 1 = 8 Marks]

VOLTAGE DROP

Question 12.
In a 400V, three phase, non-domestic installation, a 30A three-phase Oven which operates continuously, is supplied from a subcircuit originating at a distribution board.

The Consumers mains cables are:
* V90 multicore insulated and sheathed cables with circular copper conductors
* installed in the ground to a depth of 600mm installed in a wiring enclosure
* soil temperature is 15 degrees C
* protected by HRC fuses
* not installed with other cables

The submain and final sub-circuit cables are:
* V90 single-core insulated and sheathed cables with circular copper conductors
* installed in the ground to a depth of 600mm installed in a wiring enclosure
* soil temperature is 15 degrees C
* protected by HRC fuses
* not installed with other cables

The circuit details are:

Consumer Mains - multi-core cable		
	MD current	65A
	Length	45m
	Size	25mm^2
Sub-mains - Single Double Insulated		
	MD current	42A
	Length	30m
	Size	10mm^2
Final sub-circuit - Single Double Insulated		
	Length	60m
	Size	6mm^2

The Consumers Mains and Submains are at their normal operating temperatures. The final subcircuit cables are operating at 60"C.
Calculate the total voltage drop from the point of supply to the oven terminals.

All calculations including the final answer must be completed to a maximum of **two decimal places.**

All relevant table details, calculations and units must be shown to obtain full marks.

Cable	Table	Column	Vc	Calculation	Vd
Consumer Mains					
Sub-mains					
Final Sub-circuit					

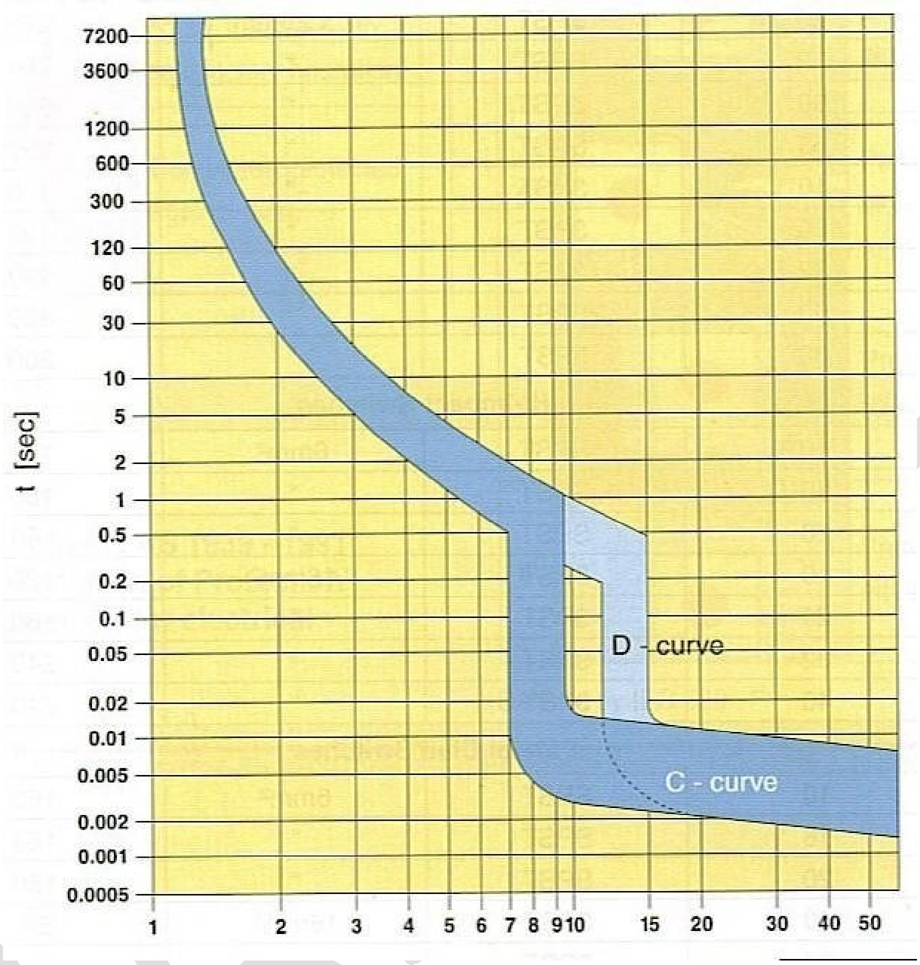
Answer Total Vd:	

[1+1+1+1+1+1+1+1+1+1 = 10 Marks]

OVERLOAD AND SHORT CIRCUIT CALCULATIONS

Question 13.

What are the minimum and maximum tripping times for a 25A Type C miniature over-current circuit breaker which is subjected to an over-current of 100A?



Overcurrent divided by MCB Current Rating

Overcurrent divided by MCB current rating:			
Minimum time:		**Maximum time:**	

[1 + 1 + 1 = 3 Marks]

OVERLOAD AND SHORT CIRCUIT CALCULATIONS

Question 14.

The main switchboard of a 400/230V industrial installation is directly supplied from a 500KVA transformer which has a prospective fault current of 13,900A per phase.

Submains supply a distribution board from the main switchboard.

The following information is known:

Impedance of the Consumer Mains = 0.00450Ω
Impedance of the Sub-mains cables = 0.02000Ω

Determine:

(i) the transformer impedance;

(ii) the prospective fault current at the main switchboard between phase and earth; and

(iii) the prospective fault current at the distribution board between phase and earth.

Work impedances to 5 decimal places.

All calculations must be shown to obtain full marks.

Transformer Impedance:	
MainSw/Bd:	
Dist/Bd:	

[(2+1) + (2+1) + (2+1) = 9 Marks]

RESIDUAL CURRENT DEVICES

Question 15.
A 30mA Residual Current Device is to be installed to protect two (2) circuits of 10A socket outlets with a total maximum demand of 15A. Each circuit is protected by a 16A circuit breaker.

State the minimum current rating of the Residual Current Device.

[3 Marks]

MOTORS AND STARTERS

Question 16.

In this question, circle the letter in front of the statement that correctly completes the following sentence.

Control circuits associated with the operation of fire pump motors shall be arranged so that the active conductor of the control circuit is directly connected to the coil of the operating device within the………………

A. main switchboard
B. fire control panel
C. starter
D. motor junction box

[2 Marks]

AS/NZS 4836:2011

Question 17.

This question relates to AS/NZS 4836:2011.

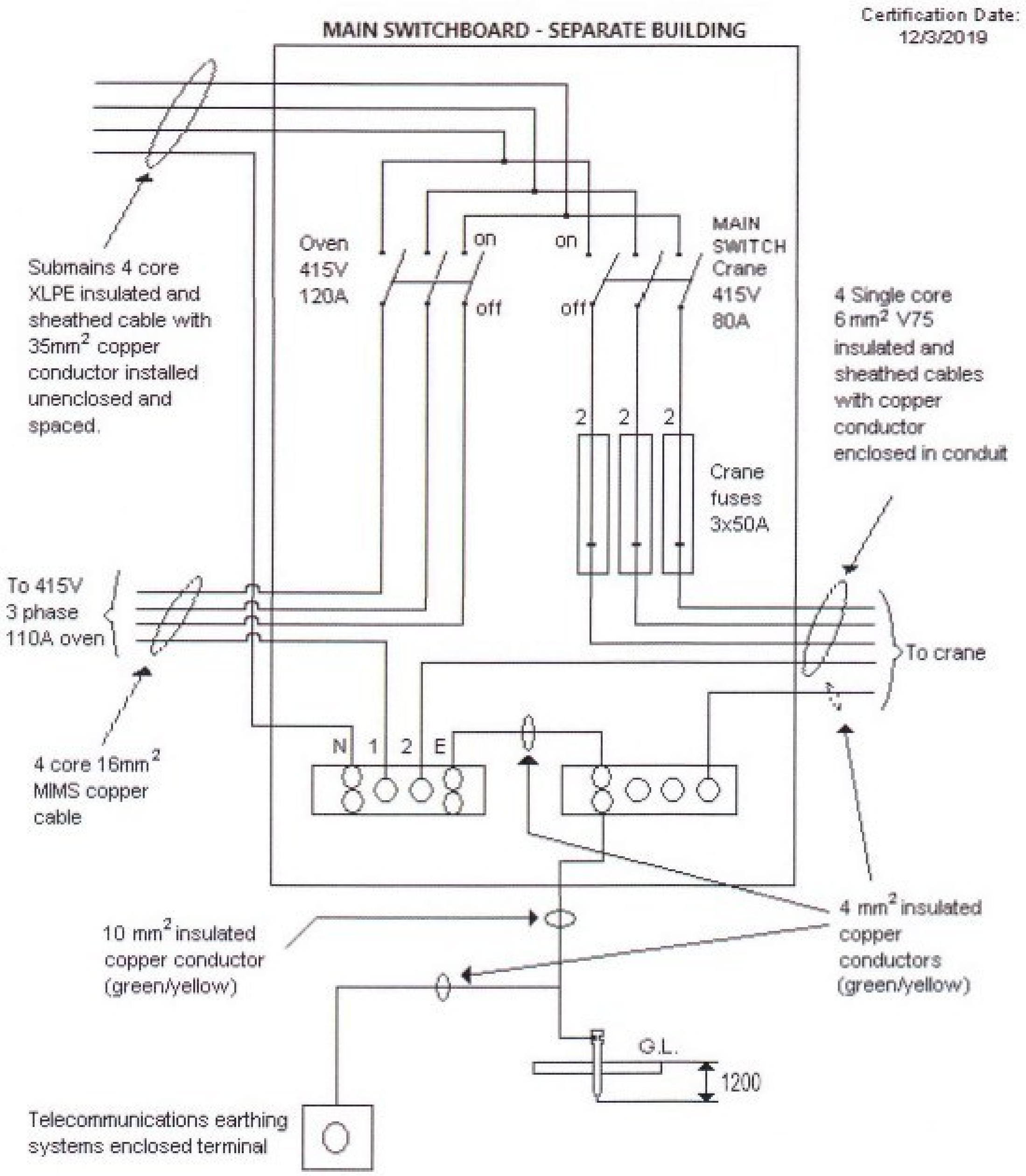
Name 2 (two) examples of areas of reduced mobility.

1)

2)

Standard Clause Number: [1 + 1 + 2 = 4 Marks]

INSTALLATION DEFECTS - NON DOMESTIC



Question 18.
The drawing above shows a SWITCHBOARD located in a separate building of an Industrial Complex. It supplies an electric oven rated at 110A per phase and a crane rated at 40A per phase.

The SWITCHBOARD is supplied by a submain from a MAIN SWITCHBOARD located in another building.

The multi-core MIMS cables are installed spaced from the wall.

Assume the MIMS cables are earthed in accordance with the Wiring Rules.

All fuses are HRC type. The drawing contains contraventions to the Wiring Rules.

The Supply Authority has provided Short Circuit Protection of the point of Supply on the Consumers mains.

Complete the table on the following page.

Question 18. Continued.

Use the diagram on the previous page.

List **FIVE different defects** together with the contravened Wiring Rules Clause/Table number in the table provided below.

Note: Only the first five defects will be considered.

DEFECT DETAILS	WIRING RULE CLAUSE/TABLE No.

5 x (2 + 1) = 15 Marks]